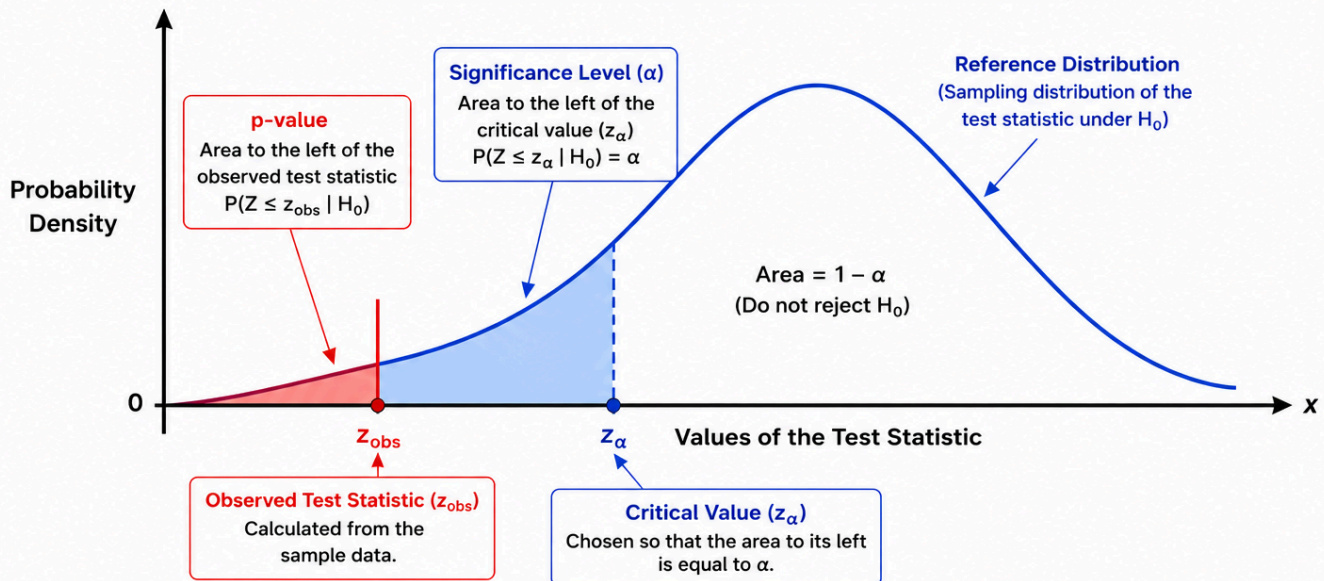


Types of Tailed Tests

Left-Tailed Test

H_0 : parameter $\geq$ value₀ vs. H_a : parameter $<$ value₀
(Reject H_0 for small values of the test statistic)



Decision Rule (Left-Tailed Test)

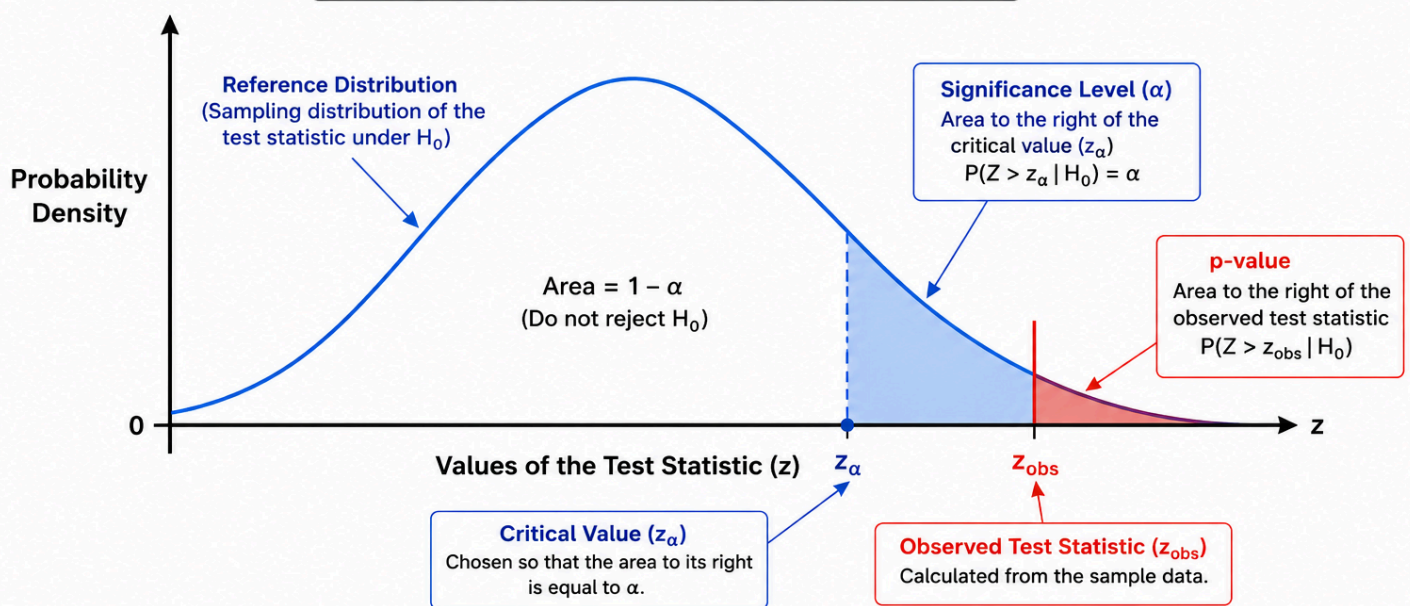
- Reject H_0 if $z_{obs} < z_\alpha$ (equivalently, if **p-value** $<$ α)
- Otherwise, do not reject H_0 .

Key Takeaway

α is the pre-chosen threshold (area in the left tail).
p-value is the area to the left of what we observed.

Right-Tailed Test

H_0 : parameter $\leq$ value₀ vs. H_a : parameter $>$ value₀
(Reject H_0 for large values of the test statistic)



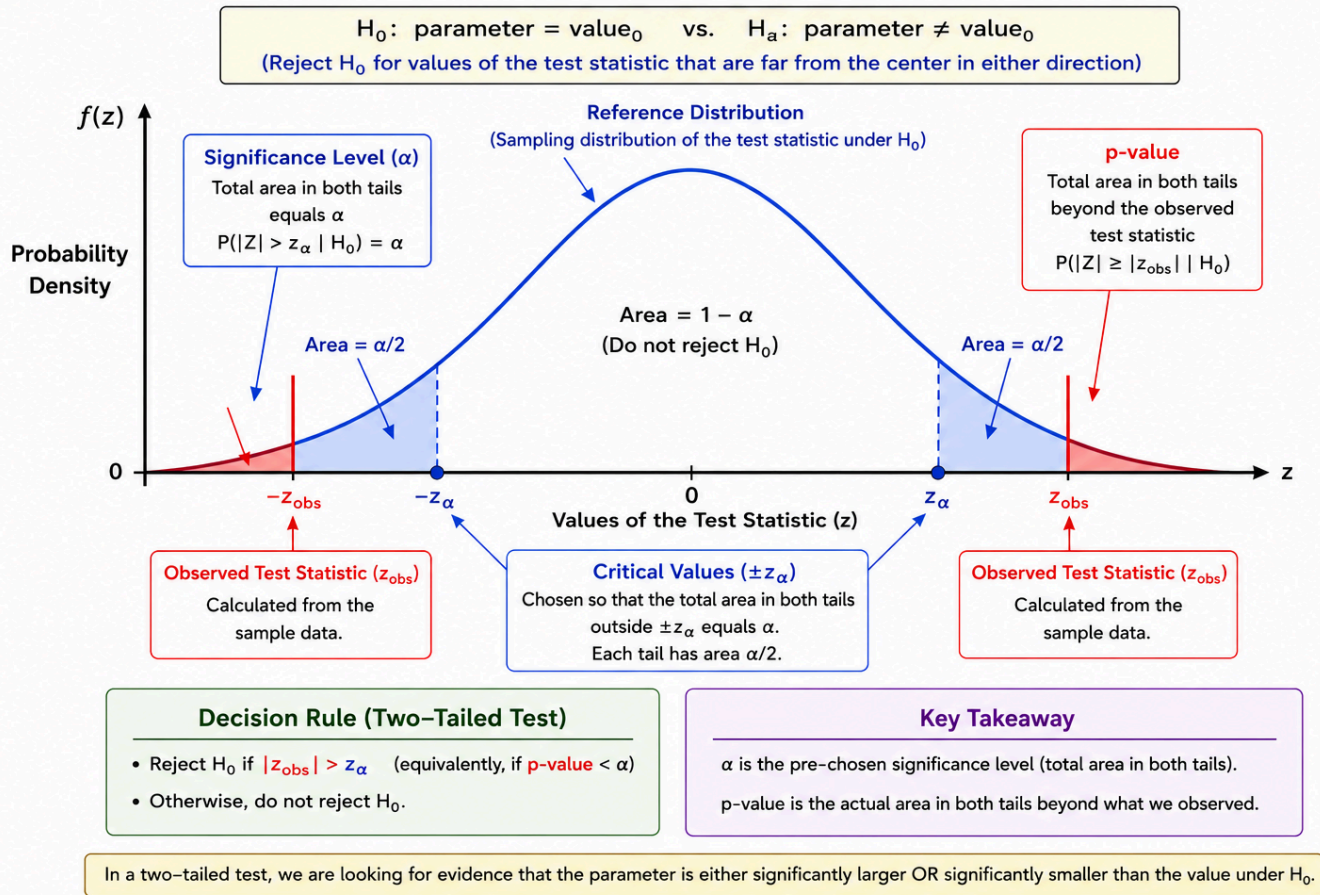
Decision Rule (Right-Tailed Test)

- Reject H_0 if $z_{obs} > z_\alpha$ (equivalently, if **p-value** $<$ α)
- Otherwise, do not reject H_0 .

Key Takeaway

α is the pre-chosen threshold (area in the right tail).
p-value is the area to the right of what we observed.

Two-Tailed Test



Note:

These Images are generated by ChatGPT, using below prompts.

Image Prompts

Prompt for Left Tailed Test

Generate an image clearly demonstrating p-value and significance-level on Reference Distribution for Left tailed test. Correctly label X-axis and Y-Axis.

Prompt for Right Tailed Test

Generate a image clearly demonstrating p-value and significance-level on Reference Distribution for right tailed test. Correctly label X-axis and Y-Axis.

Prompt for Two Tailed Test

Generate an image clearly demonstrating p-value and significance-level on Reference Distribution for Two tailed test. Correctly label X-axis and Y-Axis.